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| Tool Version     : Vivado v.2025.2 (win64) Build 6299465 Fri Nov 14 19:35:11 GMT 2025
| Date             : Wed Jul  1 22:38:49 2026
| Host             : AYODELE running 64-bit major release  (build 9200)
| Command          : report_power -file Matrix_Accelerator_power_routed.rpt -pb Matrix_Accelerator_power_summary_routed.pb -rpx Matrix_Accelerator_power_routed.rpx
| Design           : Matrix_Accelerator
| Device           : xc7k70tfbv676-1
| Design State     : routed
| Grade            : commercial
| Process          : typical
| Characterization : Production
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```

Power Report

Table of Contents
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1. Summary

Total On-Chip Power (W)	0.089
Design Power Budget (W)	Unspecified*
Power Budget Margin (W)	NA
Dynamic (W)	0.008
Device Static (W)	0.081
Effective TJA (C/W)	1.9
Max Ambient (C)	84.8
Junction Temperature (C)	25.2
Confidence Level	Low
Setting File	---
Simulation Activity File	---
Design Nets Matched	NA

* Specify Design Power Budget using, set_operating_conditions -design_power_budget <value in Watts>

1.1 On-Chip Components

On-Chip	Power (W)	Used	Available	Utilization (%)
Clocks	0.001	3	---	---
Slice Logic	<0.001	345	---	---
LUT as Logic	<0.001	83	41000	0.20

CARRY4	<0.001	8	10250	0.08
Register	<0.001	226	82000	0.28
Others	0.000	18	---	---
Signals	0.002	476	---	---
DSPs	0.004	4	240	1.67
I/O	<0.001	154	300	51.33
Static Power	0.081			
Total	0.089			

1.2 Power Supply Summary

Source	Voltage (V)	Total (A)	Dynamic (A)	Static (A)	Powerup (A)	Budget (A)	Margin (A)
Vccint	1.000	0.029	0.008	0.022	NA	Unspecified	NA
Vccaux	1.800	0.012	0.000	0.012	NA	Unspecified	NA
Vcco33	3.300	0.000	0.000	0.000	NA	Unspecified	NA
Vcco25	2.500	0.000	0.000	0.000	NA	Unspecified	NA

```
| Vcco18    |       1.800 |     0.001 |       0.000 |      0.001 |
    NA    | Unspecified | NA         |
| Vcco15    |       1.500 |     0.000 |       0.000 |      0.000 |
    NA    | Unspecified | NA         |
| Vcco135   |       1.350 |     0.000 |       0.000 |      0.000 |
    NA    | Unspecified | NA         |
| Vcco12    |       1.200 |     0.000 |       0.000 |      0.000 |
    NA    | Unspecified | NA         |
| Vccaux_io |       1.800 |     0.000 |       0.000 |      0.000 |
    NA    | Unspecified | NA         |
| Vccbram   |       1.000 |     0.000 |       0.000 |      0.000 |
    NA    | Unspecified | NA         |
| MGTAVcc   |       1.000 |     0.000 |       0.000 |      0.000 |
    NA    | Unspecified | NA         |
| MGTAVtt   |       1.200 |     0.000 |       0.000 |      0.000 |
    NA    | Unspecified | NA         |
| MGTVccaux |       1.800 |     0.000 |       0.000 |      0.000 |
    NA    | Unspecified | NA         |
| Vccadc    |       1.800 |     0.020 |       0.000 |      0.020 |
    NA    | Unspecified | NA         |
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1.3 Confidence Level
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| User Input Data             | Confidence | Details
                                | Action

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| Design implementation state | High        | Design is routed                                |                                                                                |
| Clock nodes activity        | High        | User specified more than 95% of clocks          |                                                                                |
| I/O nodes activity          | Low         | More than 75% of inputs are missing user specification | Provide missing input activity with simulation results or by editing the "By Resource Type -> I/Os" view   |
| Internal nodes activity     | Medium      | User specified less than 25% of internal nodes  | Provide missing internal nodes activity with simulation results or by editing the "By Resource Type" views |
| Device models               | High        | Device models are Production                    |                                                                                |
|                             |             |                                                 |                                                                                |
| Overall confidence level    | Low         |                                                 |                                                                                |
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```

2. Settings

2.1 Environment

Ambient Temp (C)	25.0
ThetaJA (C/W)	1.9
Airflow (LFM)	250
Heat Sink	medium (Medium Profile)
ThetaSA (C/W)	3.4
Board Selection	medium (10"x10")
# of Board Layers	12to15 (12 to 15 Layers)
Board Temperature (C)	25.0

2.2 Clock Constraints

Clock	Domain	Constraint (ns)
Clock	Clock	10.0

3. Detailed Reports

3.1 By Hierarchy

Name	Power (W)

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| Matrix_Accelerator
      |     0.008 |
|   SA_Inst
      |     0.005 |
|
Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_Inst
|     0.001 |
|       MAC_Unit_Inst
      |     0.001 |
|
Gen_PE_row[0].Gen_PE_col[1].Gen_Interior_Col.Gen_Top_Edge.PE_Inst
|     0.001 |
|       MAC_Unit_Inst
      |     0.001 |
|
Gen_PE_row[1].Gen_PE_col[0].Gen_Left_Edge.Gen_Left_Interior.PE_Inst
|     0.001 |
|       MAC_Unit_Inst
      |     0.001 |
|
Gen_PE_row[1].Gen_PE_col[1].Gen_Interior_Col.Gen_Interior.PE_Inst
|     0.001 |
|       MAC_Unit_Inst
      |     0.001 |
+-----------------------------------------------------------------
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```